



Parul University

Laboratory Manual

FACULTY OF ENGINEERING AND TECHNOLOGY
BACHELOR OF TECHNOLOGY

COMPUTER NETWORKS LABORATORY
(303105256)

IV SEMESTER

Computer Science & Engineering Department



Laboratory Manual
Session 2025-26

COMPUTER NETWORKS PRACTICAL BOOK
COMPUTER SCIENCE & ENGINEERING DEPARTMENT

PREFACE

It gives us immense pleasure to present the first edition of the **COMPUTER NETWORKS** Practical Book for the B.Tech . 4rd **semester** students for **PARUL UNIVERSITY**.

The **CN** theory and laboratory courses at **PARUL UNIVERSITY**, **WAGHODIA**, **VADODARA** are designed in such a way that students develop the basic understanding of the subject in the theory classes and then try their hands on the experiments to realize the various implementations of problems learnt during the theoretical sessions. The main objective of the **CN** laboratory course is: Learning **CN** through Experimentations. All the experiments are designed to illustrate various problems in different areas of **CN** and also to expose the students to various uses.

The objective of this **CN** Practical Book is to provide a comprehensive source for all the experiments included in the **CN** laboratory course. It explains all the aspects related to every experiment such as: basic underlying concept and how to analyze a problem. It also gives sufficient information on how to interpret and discuss the obtained results.

We acknowledge the authors and publishers of all the books which we have consulted while developing this Practical book. Hopefully this **CN** Practical Book will serve the purpose for which it has been developed.

INSTRUCTIONS TO STUDENTS

1. The main objective of the **CN** laboratory is: Learning through the Experimentation. All the experiments are designed to illustrate various problems in different areas of **CN** and also to expose the students to various problems and their uses.
2. Be prompt in arriving to the laboratory and always come well prepared for the practical.
3. Every student should have his/her individual copy of the **CN** Practical Book.
4. Every student have to prepare the notebooks specifically reserved for the **CN** practical work: "CN Practical Book"
5. Every student has to necessarily bring his/her **CN** Practical Book, **CN** Practical Class Notebook and **CN** Practical Final Notebook.
6. Finally find the output of the experiments along with the problem and note results in the **CN** Practical Notebook.
7. The grades for the **CN** practical course work will be awarded based on our performance in the laboratory, regularity, recording of experiments in the **CN** Practical Final Notebook, lab quiz, regular viva-voce and end-term examination.

CERTIFICATE

This is to certify that

Mr. **Naveen Sankhla** with enrolment no. **2403051050348** has successfully completed his laboratory experiments in the **COMPUTER NETWORKS (303105256)** from the department of **COMPUTER SCIENCE & ENGINEERING** during the academic year 2025-2026.



Date of Submission:.....

Staff In charge:.....

Head Of Department:.....

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2.	Experiments of Packet capture tool: Wireshark To understand the features of wireshark as a packet capture tool and understand encapsulation of information. Also study effect of few network commands						
3.	To study behaviour of generic devices used for networking: (CISCO PACKET TRACER) Design a simple network with multiple nodes and connect via generic devices available in library. Perform simulation and trace communication behaviour of specified network devices. 1: Use only HUB to design a small network having 4 to 6 hosts 2: Use only a Switch to design a small network with 4 to 6 hosts. 3: Use both the device (HUB and SWITCH) for a network and find out functioning difference between switch and hub.						
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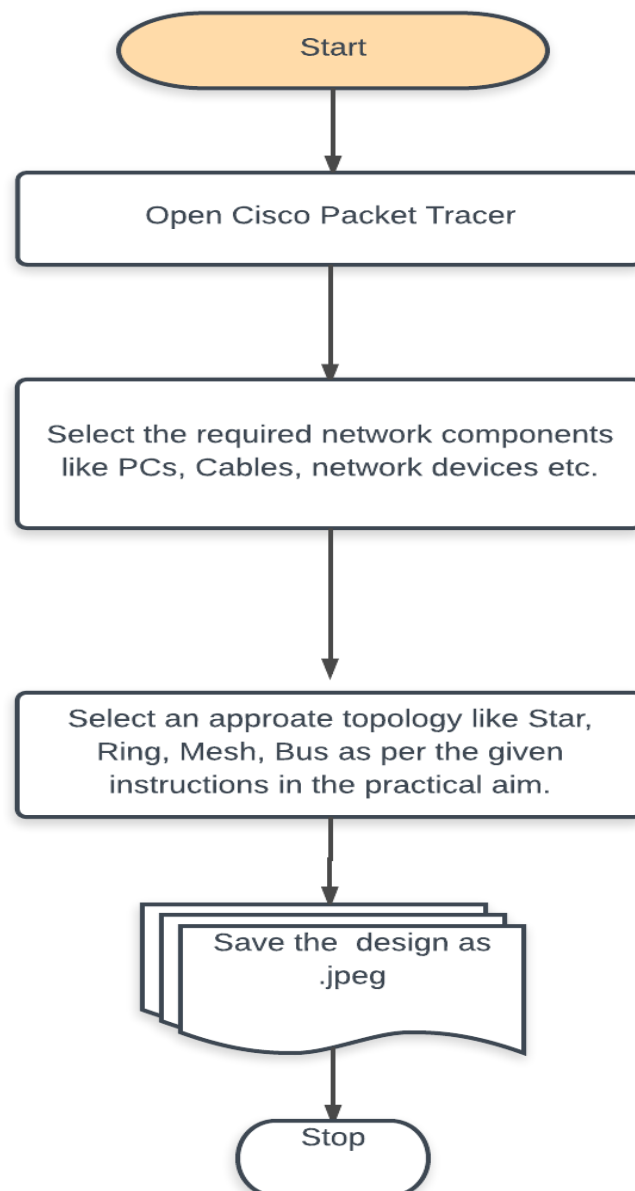
EXPERIMENT NO: 1

AIM: Experiments on Simulation Tools: (CISCO PACKET TRACER): To understand environment of CISCO PACKET TRACER to design simple network and perform experiments.

1.1 Purpose:

The main objective of the proposed experiment is to give exposure of numerous simulation devices to students such that student can actually design various network topologies in real-time.

1.2 Logical Flow of Practical



Description:

Downloading Cisco Packet Tracer

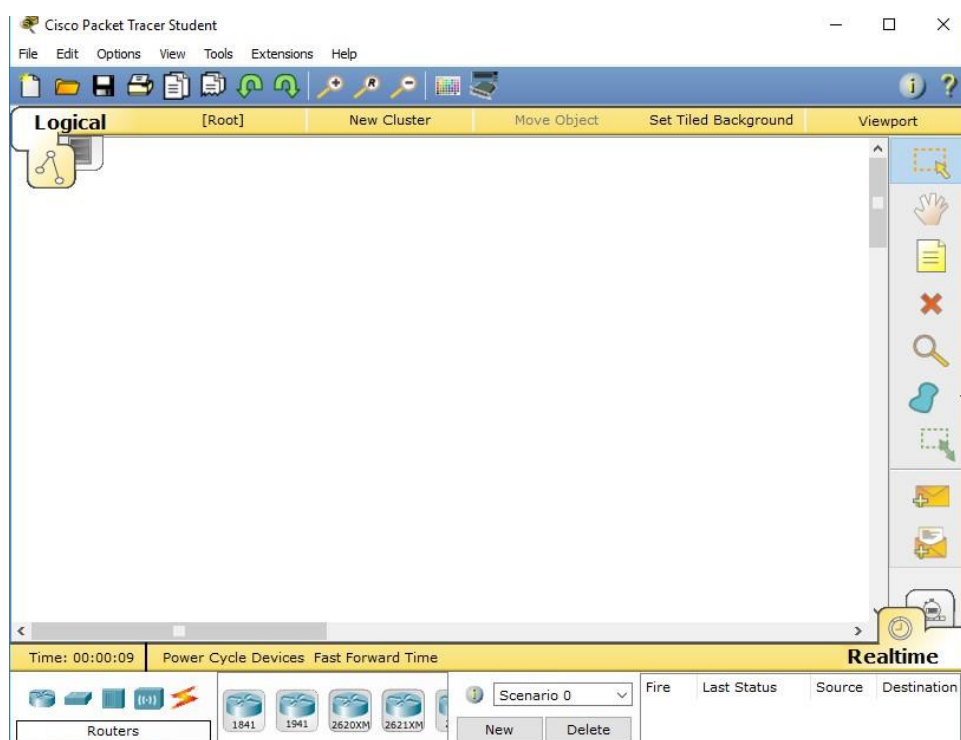
Cisco Packet Tracer can be downloaded directly from the Cisco Networking Academy (NetAcad) portal. The software is available free of cost to registered NetAcad users and supports multiple operating systems including Windows, macOS, and Linux.

To initiate the download, users must log in with their NetAcad credentials and navigate to the official Lab Downloads section. From there, the latest version of Packet Tracer can be selected according to the operating system in use. The download provides the installer file required for subsequent installation and setup.

To obtain and install your copy of Cisco Packet Tracer, please follow the instructions from the link below: <https://www.netacad.com/resources/lab-downloads>

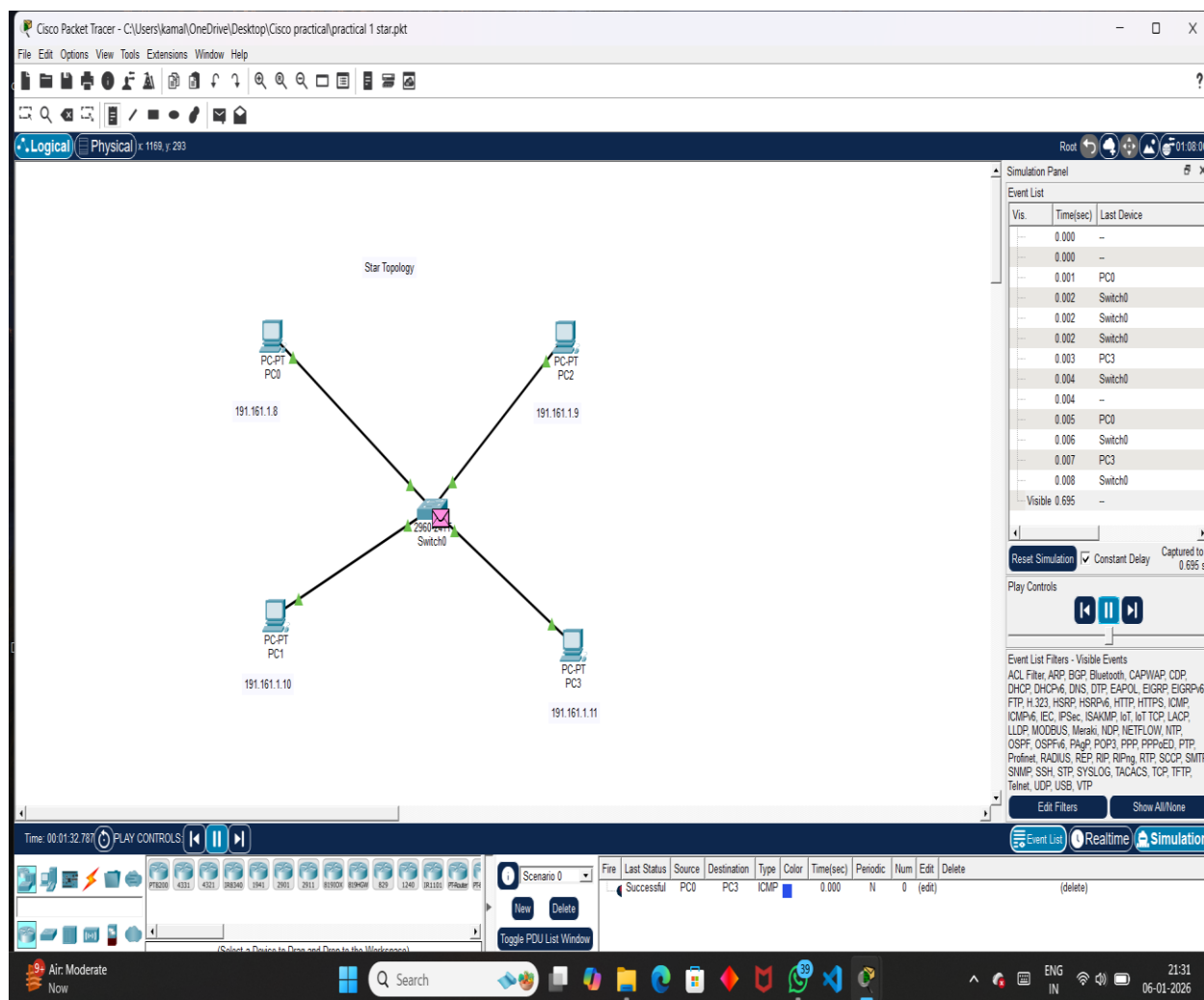
1.3 TOOLS SOFTWARE

Cisco Packet Tracer



1.4 Star Topology

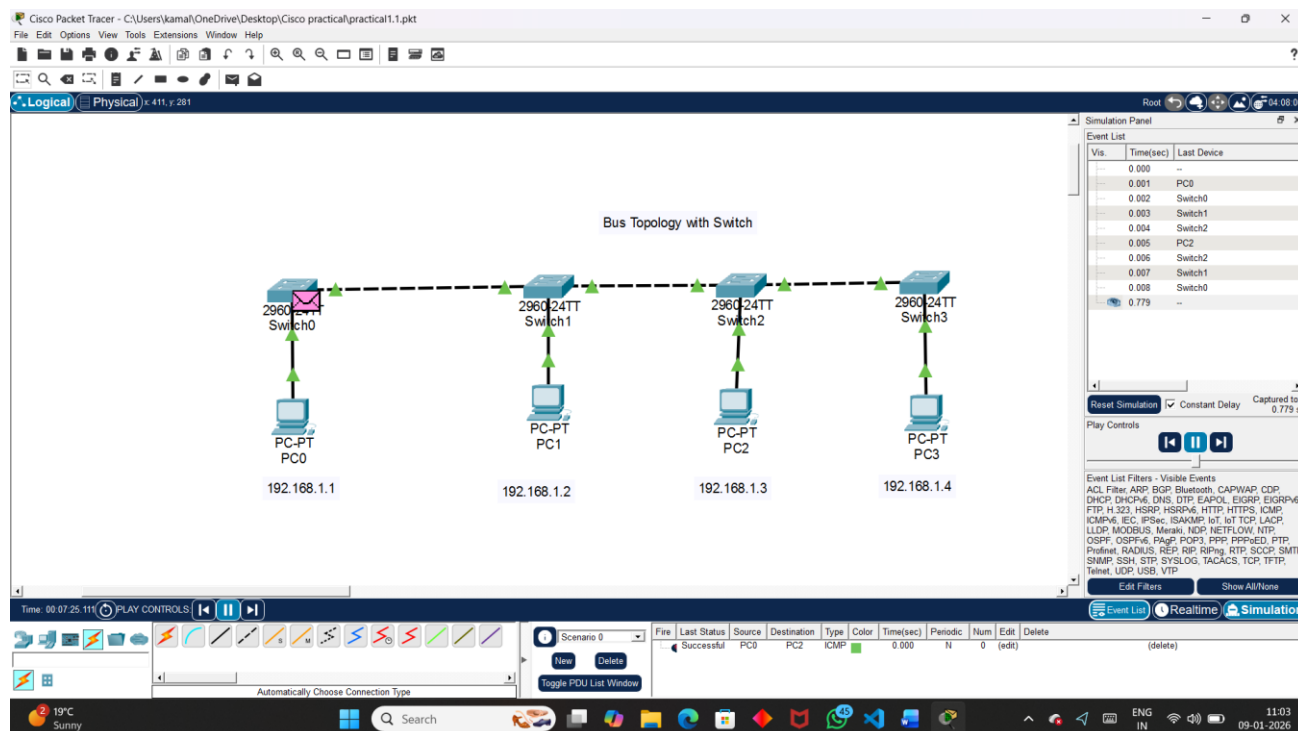
In a **star topology**, all devices are connected to a central hub or switch, which acts as the main communication point.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	191.161.1.8	f0/1	255.255.0.0	-	-
2	PC1	191.161.1.10	f0/2	255.255.0.0	-	-
3	PC2	191.161.1.9	f0/3	255.255.0.0	-	-
4	PC3	191.161.1.11	f0/4	255.255.0.0	-	-

1.5 Bus Topology

In a bus topology, all devices share a single communication line (the bus), and data travels along this backbone.

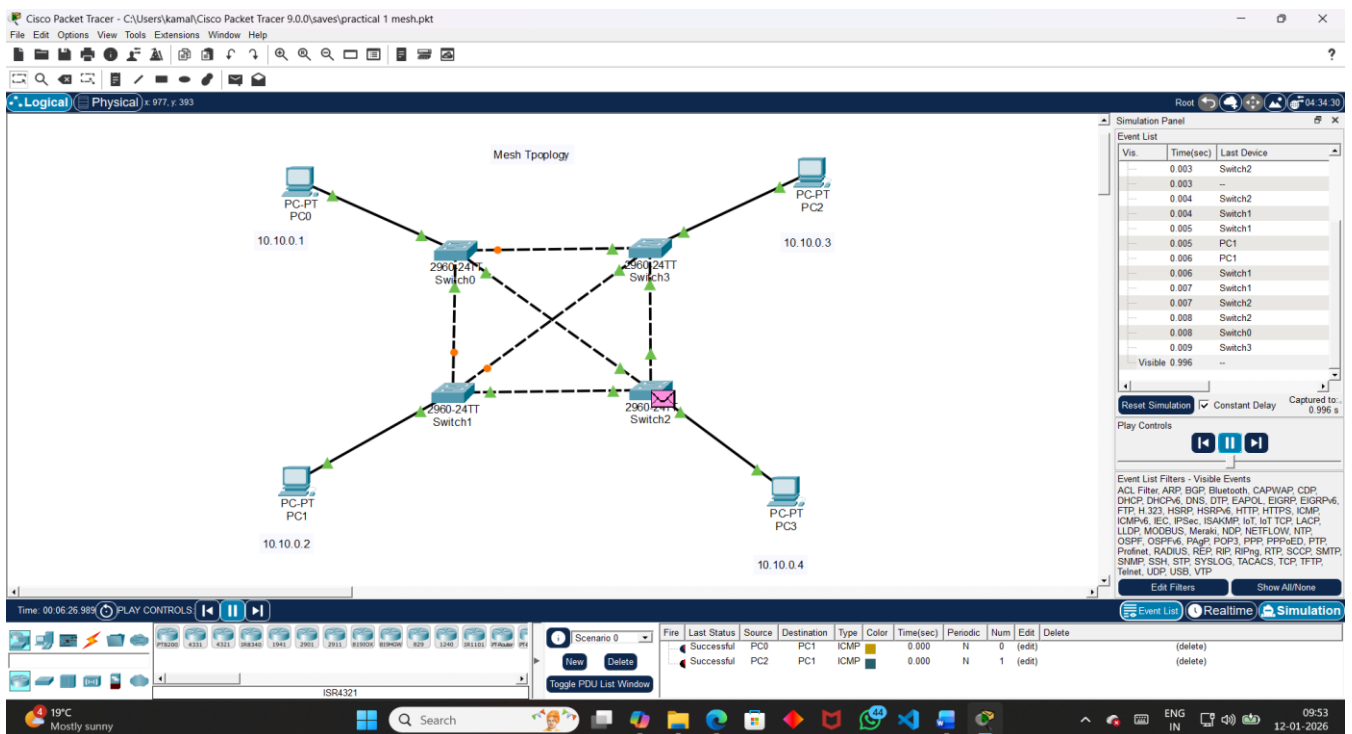


S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	192.168.1.1	f0/1	255.255.255.0	-	-
2	PC1	192.168.1.2	f0/2	255.255.255.0	-	-
3	PC2	192.168.1.3	f0/3	255.255.255.0	-	-
4	PC3	192.168.1.4	f0/4	255.255.255.0	-	-
5	2960-24TT Switch0	-	Fa0/0	-	-	-
6	2960-24TT Switch1	-	Fa0/1	-	-	-
7	2960-24TT Switch2	-	Fa0/2	-	-	-
8	2960-24TT Switch3	-	Fa0/3	-	-	-

1.6 Mesh Topology

In a mesh topology, every device is directly connected to every other device, creating multiple communication paths.

This design ensures high reliability and fault tolerance, but it requires more cabling and setup effort.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	10.10.0.1	f0/1	255.0.0.0	0.0.0.0	0.0.0.0
2	PC1	10.10.0.2	f0/2	255.0.0.0	0.0.0.0	0.0.0.0
3	PC2	10.10.0.3	f0/3	255.0.0.0	0.0.0.0	0.0.0.0
4	PC3	10.10.0.4	f0/4	255.0.0.0	0.0.0.0	0.0.0.0
5	2960-24TT Switch0	-	Fa0/0	-	-	-
6	2960-24TT Switch1	-	Fa0/1	-	-	-
7	2960-24TT Switch2	-	Fa0/2	-	-	-
8	2960-24TT Switch3	-	Fa0/3	-	-	-

CONCLUSION:

Here we have studied that, How to install the Cisco Packet Tracer and design some simple network with the help of Cisco Packet Tracer which was done successfully.

EXPERIMENT NO: 3

Aim : To study behaviour of generic devices used for networking: (CISCO PACKET TRACER)

Design a simple network with multiple nodes and connect via generic devices available in library. Perform simulation and trace communication behaviour of specified network devices.

1: Use only HUB to design a small network having 4 to 6 hosts

2: Use only a Switch to design a small network with 4 to 6 hosts. 3: Use both the device (HUB and SWITCH) for a network and find out functioning difference between switch and hub

Purpose / Objective

- To understand how a Hub operates at the physical layer by broadcasting signals to all connected hosts.
- To observe how a Switch operates at the data link layer by forwarding frames only to the intended destination.
- To compare the functioning of Hub and Switch when used together in a network.
- To analyze communication efficiency, collision domains, and traffic management in different setups.

Logical Flow of Practical

1. Network with Hub only
 - Create 4–6 hosts (PCs).
 - Connect all hosts to a Hub using straight-through Ethernet cables.
 - Assign IP addresses in the same subnet.
 - Simulate communication (ping) and observe broadcast behavior.
2. Network with Switch only
 - Create 4–6 hosts (PCs).
 - Connect all hosts to a Switch using straight-through Ethernet cables.
 - Assign IP addresses in the same subnet.
 - Simulate communication (ping) and observe selective forwarding behavior.
3. Network with Hub and Switch combined

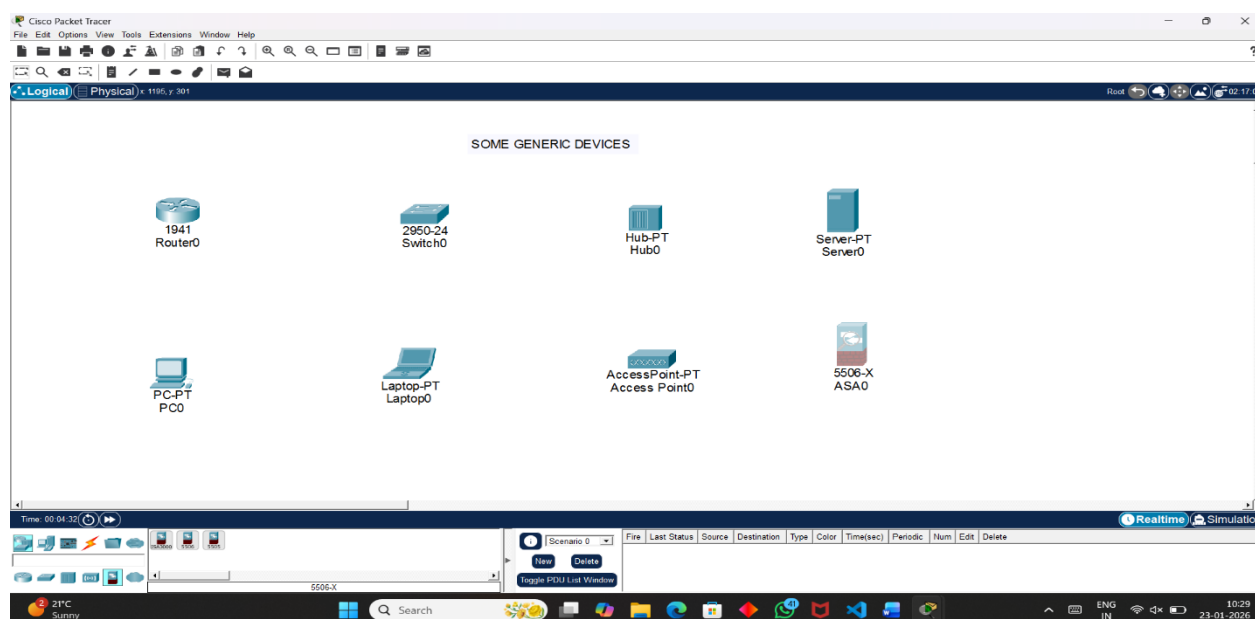
- Connect 2–3 hosts to a Hub and 2–3 hosts to a Switch.
- Interconnect the Hub and Switch.
- Assign IP addresses in the same subnet.
- Simulate communication across devices and compare how Hub and Switch handle traffic.

Description

In this practical, three different network topologies are designed using Cisco Packet Tracer to study the behavior of generic networking devices: Hub and Switch.

- In the Hub-based network, all data frames are broadcast to every connected host, leading to collisions and inefficient communication.
- In the Switch-based network, the switch learns MAC addresses and forwards frames only to the destination host, ensuring efficient communication without collisions.
- In the combined Hub-Switch network, the difference in functionality is clearly observed: hosts connected to the Hub experience broadcasts, while hosts connected to the Switch communicate efficiently.

Cisco Packet Tracer is perfect for studying how different networking devices behave in simulated environments. Here are **8 commonly used generic devices** you'll find in Packet Tracer:



1. Router

A router operates at the network layer (Layer 3) of the OSI model and connects multiple networks together.

It uses IP addresses to determine the best path for data and supports routing protocols, NAT, and access control.

In Packet Tracer, routers are used to simulate real-world WAN and inter-LAN communication.



2. Switch

A switch works at the data link layer (Layer 2) and connects devices within the same local network. It maintains a MAC address table to forward frames only to the intended destination, improving efficiency.

Packet Tracer switches help in learning VLANs, trunking, and basic network segmentation.



3. Hub

A hub is a simple physical layer (Layer 1) device with no decision-making capability.

Any data received on one port is sent to all other ports, causing collisions and poor performance.

Though obsolete in real networks, hubs are useful for understanding basic data transmission concepts.



4. Wireless Router

A wireless router combines the functions of a router, switch, and access point. It routes traffic between networks while also providing Wi-Fi connectivity to wireless devices. In Packet Tracer, it is commonly used in home or small office network simulations.



5. Access Point

An access point connects wireless devices to a wired LAN without performing routing. It acts as a bridge between wired and wireless networks, extending Wi-Fi coverage. Packet Tracer uses access points to demonstrate wireless standards and coverage expansion.



8. Server

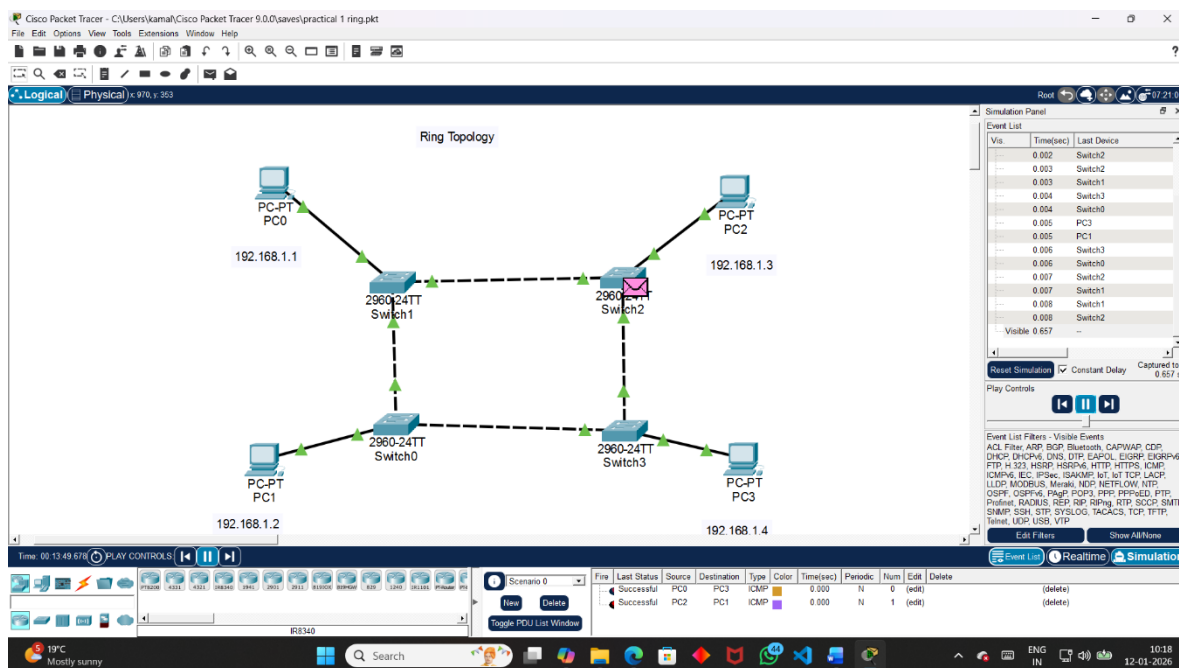
A server is a dedicated device that provides services to other devices on the network. It can host services like DNS, DHCP, HTTP, FTP, and Email within Packet Tracer. Servers are crucial for understanding centralized network management and service delivery.



- **Ring Topology**

In a **ring topology**, each device is connected to exactly two others, forming a closed loop.

Data travels in one direction around the ring until it reaches its destination.

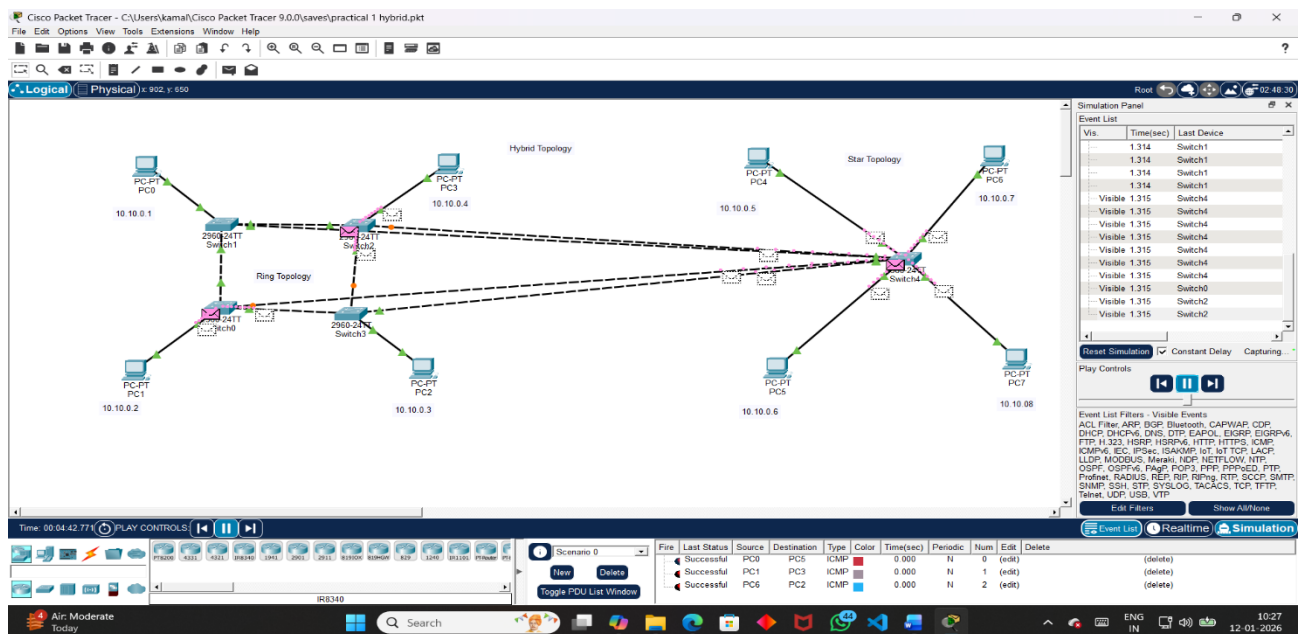


S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	192.168.1.1	f0/1	255.255.255.0	-	-
2	PC1	192.168.1.2	f0/2	255.255.255.0	-	-
3	PC2	192.168.1.3	f0/3	255.255.255.0	-	-
4	PC3	192.168.1.4	f0/4	255.255.255.0	-	-
5	2960-24TT Switch0	-	Fa0/1	-	-	-
6	2960-24TT Switch1	-	Fa0/2	-	-	-
7	2960-24TT Switch2	-	Fa0/3	-	-	-
8	2960-24TT Switch3	-	Fa0/4	-	-	-

• Hybrid Topology

A **hybrid topology** combines two or more different network topologies (like star, bus, or ring).

It leverages the strengths of each topology while minimizing their weaknesses for flexible network design.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	10.10.0.1	f0/1	255.0.0.0	-	-
2	PC1	10.10.0.2	f0/2	255.0.0.0	-	-
3	PC2	10.10.0.3	f0/3	255.0.0.0	-	-
4	PC3	10.10.0.4	f0/4	255.0.0.0	-	-
5	PC4	10.10.0.5	f0/5	255.0.0.0	-	-
6	PC5	10.10.0.6	f0/6	255.0.0.0	-	-
7	PC6	10.10.0.7	f0/7	255.0.0.0	-	-
8	PC7	10.10.0.8	f0/8	255.0.0.0	-	-
9	2960-24TT Switch0	-	Fa0/1	-	-	-
10	2960-24TT Switch1	-	Fa0/2	-	-	-
11	2960-24TT Switch2	-	Fa0/3	-	-	-
12	2960-24TT Switch3	-	Fa0/4	-	-	-
13	2960-24TT Switch4	-	Fa0/5	-	-	-

CONCLUSION:

Here we have studied that, the difference between the hub and switch and with some network with the switch and hub which was done successfully.

EXPERIMENT NO: 5

Aim : Virtual LAN: Simulate Virtual LAN configuration using CISCO Packet Tracer Simulation.

Purpose / Objective:-

- To group devices based on functions or departments (e.g., HR, Engineering, Sales) rather than their physical location.
- To ensure that sensitive data in one VLAN is not accessible to users in another VLAN without a router or Layer 3 switch.
- To reduce unnecessary network congestion by limiting the reach of broadcast frames.

Logical Flow of Practical:

Step 1: 1. Topology Design & Physical Cabling

The first step involves placing the necessary hardware onto the workspace.

- **Select Devices:** Drag and drop a Cisco Switch (e.g., 2960) and multiple End Devices (PCs or Laptops).
- **Physical Connections:** Use Copper Straight-Through cables to connect the PCs to the switch Ethernet ports.
- **Labeling:** Visually label the PCs on the workspace (e.g., "PC-0: Sales", "PC-1: HR") to avoid confusion during the CLI configuration.

Step 2: IP Address Assignment

Before configuring VLANs, the devices must have valid IP addresses to test connectivity.

- Assign static IP addresses to all PCs within the same subnet (e.g., 192.168.1.0/24).
- Baseline Test: Perform a "Ping" test between all PCs. At this stage, all pings should succeed because they are all in the default VLAN 1.

Step 3: VLAN Database Creation

Now, move into the Switch Command Line Interface (CLI) to define the virtual boundaries.

- Enter Global Configuration Mode.

- Create VLAN IDs (e.g., vlan 10) and assign them descriptive names (e.g., name Marketing).
- This step creates the "buckets" into which the ports will eventually be placed.

Step 4: Port Membership Assignment (Access Ports)

This is where the logical segmentation actually happens.

- Select specific interfaces (e.g., interface fastEthernet 0/1).
- Set the port mode to Access.
- Assign the port to a specific VLAN ID. Once assigned, that port is logically isolated from any port not in the same VLAN.

Step 5: Trunk Link Configuration (Optional/Multi-switch)

If your experiment uses more than one switch:

- Identify the link connecting the two switches.
- Configure that interface as a Trunk (switchport mode trunk).
- This allows traffic from *all* VLANs to travel across a single cable between switches.

OBSERVATION:

When all PCs are connected to the switch and assigned IP addresses in the same subnet, they can all communicate freely. By running the command `show vlan brief`, you will observe that all switch ports (FastEthernet 0/1, 0/2, etc.) are assigned to VLAN 1 by default. At this stage, the switch acts as a single broadcast domain. A broadcast message sent by one PC is received by every other PC on the switch.

DESCRIPTION:

1. Conceptual Framework

In a standard switch, all ports belong to **VLAN 1** by default. This means every device is part of the same **Broadcast Domain**. If one computer sends an ARP request, every other computer processes that request, wasting CPU cycles and bandwidth.

By creating VLANs, we are telling the switch to maintain separate **MAC Address Tables** for different groups of ports..

2. Hardware and Software Requirements

- **Software:** Cisco Packet Tracer (Version 7.0 or higher).
- **Hardware (Simulated):**
 - **1x Cisco Catalyst 2960 Switch:** Supports up to 24 FastEthernet ports.
 - **4x End Devices (PCs):** To represent different departments..

3. Detailed Execution Steps

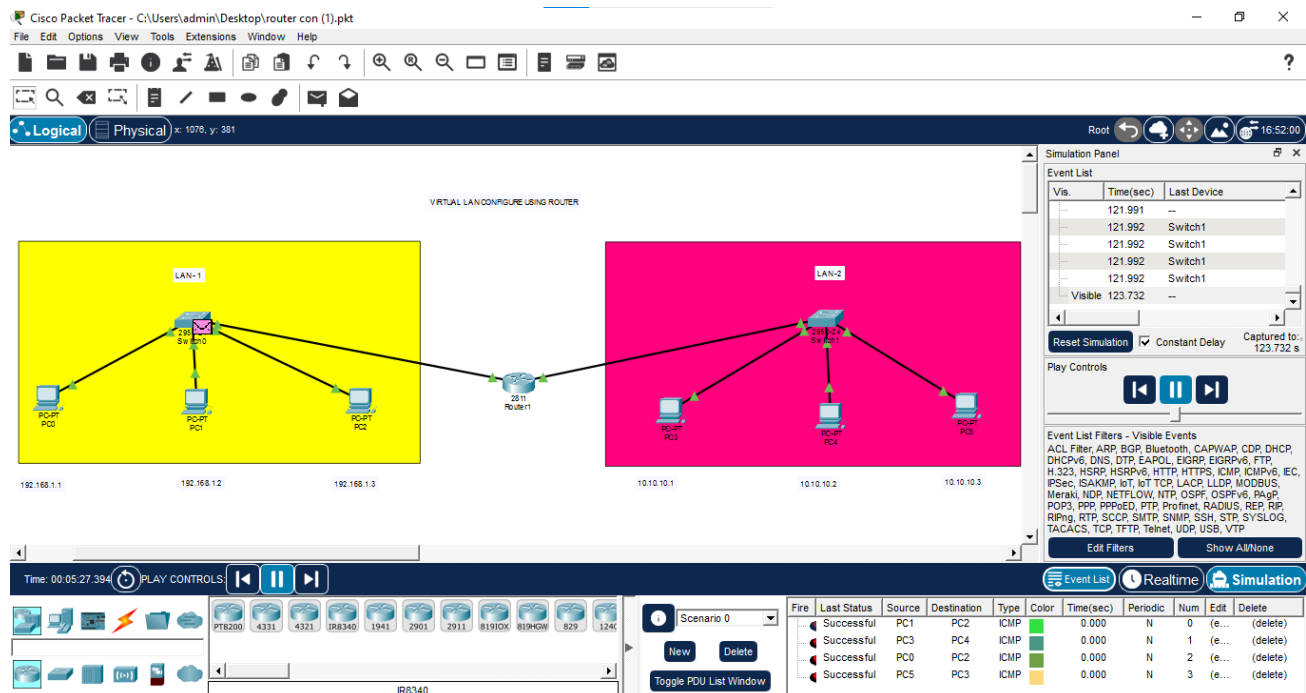
Phase A: The Physical Setup

1. Place the **2960 Switch** in the workspace.
2. Add **4 PCs** and name them according to their departments:
 - **PC0 & PC1:** Marketing Department.
 - **PC2 & PC3:** Finance Department.
 - **Connection Scheme:** PC0 -> Port Fa0/1
 - PC1 ->Port Fa0/2
 - PC2 -> Port Fa0/3
 - PC3 ->Port Fa0/4

Phase B: Logical Addressing

Assign static IP addresses to all PCs. To truly test the power of VLANs, we use the same subnet (192.168.1.0/24).

- **PC0:** 192.168.1.1
- **PC1:** 192.168.1.2
- **PC2:** 192.168.1.3
- **PC3:** 192.168.1.4
- **Subnet Mask:** 255.255.255.0



S.No	Name of Components	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	Router1	192.168.1.254	Fa0/0	255.255.255.0	-	-
2	Router1	10.10.10.254	Fa0/1	255.255.255.0	-	-
3	PC0	192.168.1.1	Wireless / NIC	255.255.255.0	192.168.1.254	-
4	PC1	192.168.1.2	Wireless / NIC	255.255.255.0	192.168.1.254	-
5	PC2	192.168.1.3	Wireless / NIC	255.255.255.0	192.168.1.254	-
6	PC3	10.10.10.1	Wireless / NIC	255.255.255.0	10.10.10.254	-
7	PC4	10.10.10.2	Wireless / NIC	255.255.255.0	10.10.10.254	-
8	PC5	10.10.10.3	Wireless / NIC	255.255.255.0	10.10.10.254	-

CONCLUSION:-

The experiment confirms that while switches provide local connectivity within a VLAN, a router is required to facilitate communication between different VLANs. This simulation provides a practical understanding of how modern corporate networks are structured to be both secure and interconnected.

EXPERIMENT NO: 6

Aim : Wireless LAN: Configuration of Wireless LAN using CISCO Packet Tracer

Purpose / Objective:-

- To understand the hardware requirements for wireless connectivity (e.g., NIC module swapping).
- To configure **SSID**, **Security Modes (WPA2)**, and **Passphrases** to protect the network.
- To verify end-to-end connectivity between wireless clients and a wired server using **Ping** and **Web Browsing**.

Logical Flow of Practical:

Step 1: Topology Design & Physical Integration

- Place network devices (Router, Server) and end-user devices (Laptop, Smartphone).
- Establish the wired backbone (Server to Router WAN).
- Hardware Modification: Swapping the Laptop's wired NIC for a wireless antenna module to enable 802.11 signal reception.

Step 2: Wired Infrastructure Configuration

- Assign a static IP to the Server to act as a fixed resource.
- Configure the Router's Internet (WAN) interface to match the server's network so they can "talk" to each other.

Step 3: Wireless Gateway Configuration (GUI)

- SSID Definition: Naming the network so clients can identify it.
- Security Implementation: Setting up WPA2-PSK encryption to protect data and restrict access.
- DHCP Service Activation: Enabling the router to automatically lease IP addresses to any wireless device that joins.

Step 4: Client Authentication and Association

- Scanning for the SSID from the Laptop/Smartphone.
- Entering the security passphrase to establish a handshake.
- DHCP Handshake: The client requests and receives an IP address, Subnet Mask, and Gateway from the router.

Step 5: Testing and Verification

- ICMP Test (Ping): Sending packets from the wireless client to the wired server to check basic connectivity.
- Application Test (HTTP): Opening a web browser on the laptop and accessing the server's IP to confirm the full network stack is working.

OBSERVATION:

Upon installing the wireless module in the Laptop, you will observe that no physical cable is needed. Once the correct **SSID** and **Password** are entered, a **dashed line** (representing a radio frequency link) appears between the client and the Router.

DESCRIPTION:

The Wireless Local Area Network (WLAN) experiment involves the creation of a flexible network environment where end devices communicate via Radio Frequency (RF) signals rather than physical cables. In this lab, we simulate a "Home or Small Office" scenario where a Wireless Router acts as the central hub.

The experiment is divided into three distinct phases:

1. **Hardware Adaptation:** Modifying legacy devices (like Laptops) to support wireless standards.
2. **Infrastructure Configuration:** Programming the Router's WAN (to talk to the internet/servers), LAN (to manage internal traffic), and Wireless (to broadcast the signal) settings.
3. **Client Integration:** Connecting mobile devices to the secure signal and verifying data flow.

STATIC ROUTING:

DESCRIPTION:

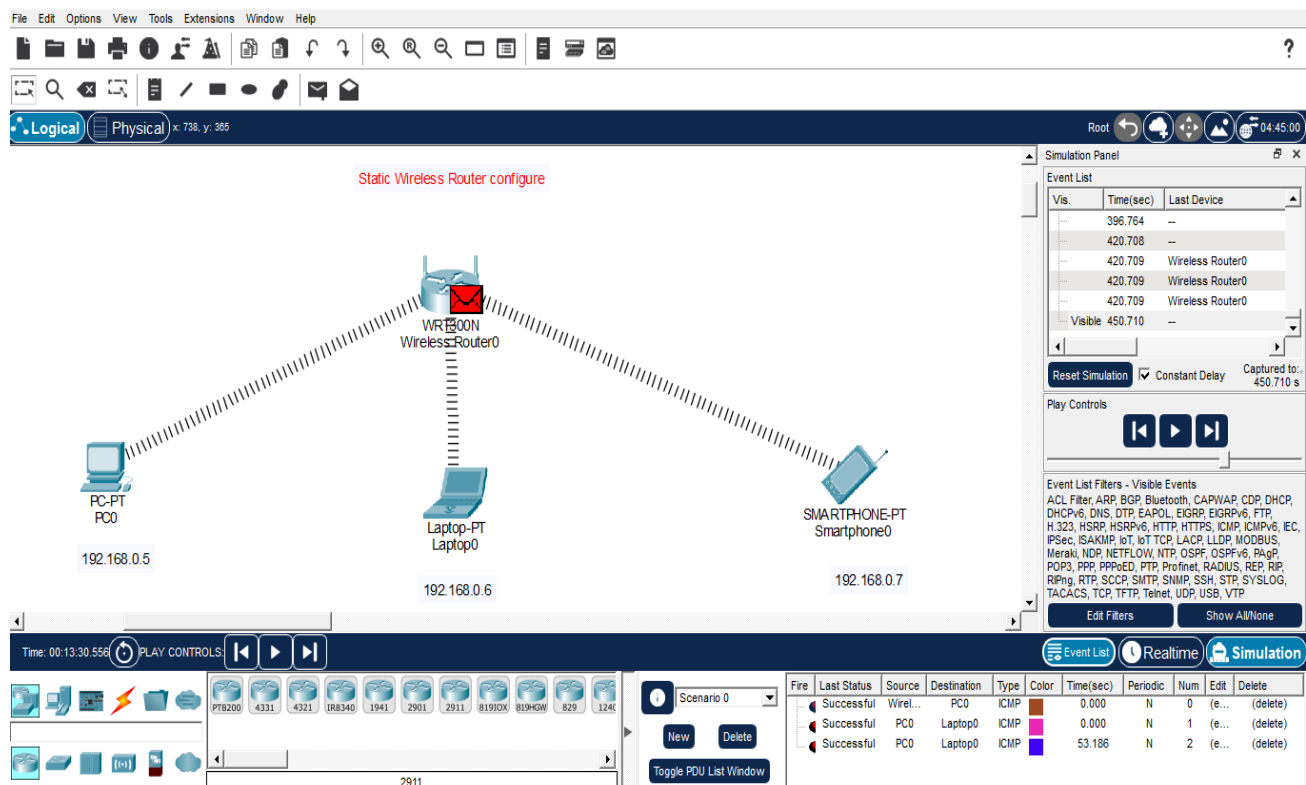
- ☐ Static routing is a method in which **routes are manually entered** into the routing table by the network administrator.
- ☐ The router **does not automatically learn or update routes** and does not use any routing protocol.
- ☐ Once configured, the route **remains fixed** until the administrator modifies or deletes it.
- ☐ It **consumes very less bandwidth, CPU, and memory** because no routing updates are exchanged.
- ☐ Static routing provides **high security** since routing information is not shared with other routers.
- ☐ It is mainly used in **small, simple, and stable networks** or as a **default route**.
- ☐ Static routing **does not handle link failures automatically**, so manual intervention is required.
- ☐ It is **not suitable for large or frequently changing networks** due to poor scalability.

Static routing is a routing technique in which routes are manually configured by the network administrator. The router does not use any routing protocol and does not automatically update routes when the network topology changes. Once a static route is set, it remains fixed until it is modified or removed manually.

Static routing uses very little bandwidth, CPU, and memory, making it suitable for small and stable networks. It provides better security because routing information is not shared with other routers. However, it is not scalable and cannot handle link failures automatically.

Static routing is a network routing method where the administrator manually defines the path for data packets. The router does not exchange routing information with other routers and cannot adjust routes automatically if the network changes.

This routing method requires minimal system resources and offers greater control and security. Static routing is best suited for small or fixed networks, but it becomes inefficient and error-prone in large networks due to the lack of automatic updates.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	WRT300N	192.168.0.1	LAN / Internal	255.255.255.0	-	-
2	PC-Pt	192.168.0.5	Wireless / NIC	255.255.255.0	192.168.0.1	-
3	Laptop-PT	192.168.0.6	Wireless0	255.255.255.0	192.168.0.1	-
4	Smartphone-PT	192.168.0.7	Wireless0	255.255.255.0	192.168.0.1	-

Dynamic Routing

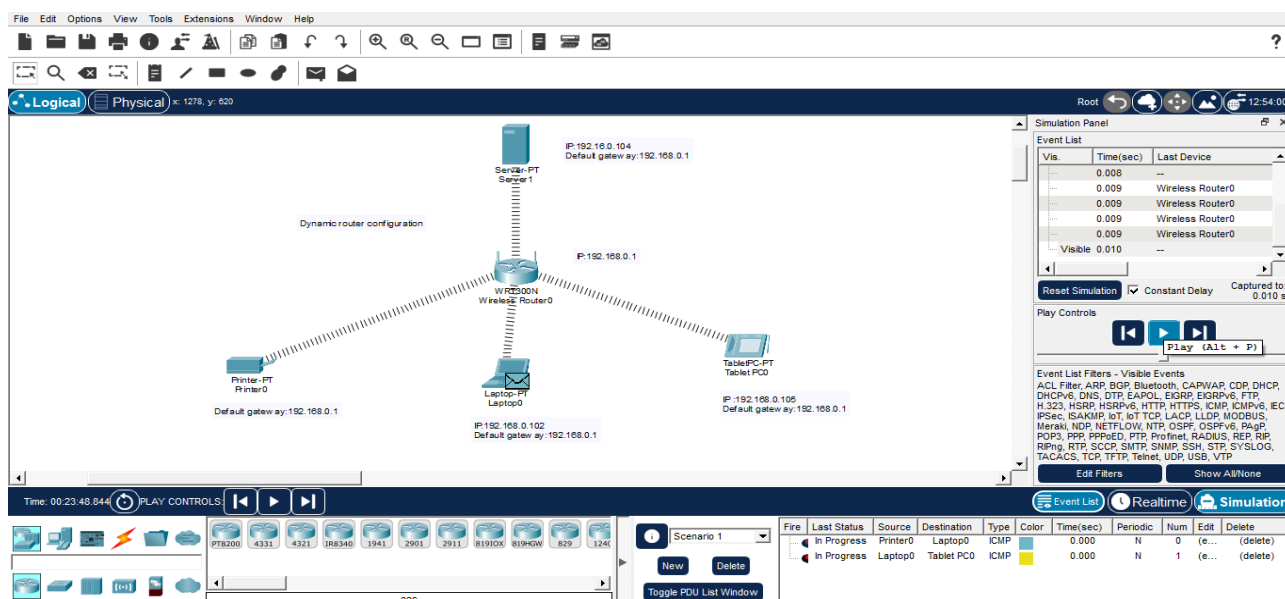
Observation Table:-

DESCRIPTION:

Dynamic routing is implemented on the Internet/WAN side of the network. This allows the wireless network to communicate with other subnets or remote servers automatically without manually configuring static routes on every device.

- **The Protocol:** Protocols like RIPv2 (Routing Information Protocol) or OSPF (Open Shortest Path First) are enabled on the routers.

- **Neighbor Discovery:** Once the routers are connected via serial or Ethernet cables, they exchange "Hello" packets to identify neighbors.
- **Route Advertising:** The core router advertises its known networks to the wireless gateway, and the wireless gateway (if it supports dynamic routing) or its upstream router advertises the path back to the location.



S.No	Name of Components	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	WRT300N	192.168.0.1	LAN / Internal	255.255.255.0	-	-
2	Server-1	192.168.0.104	FastEthernet0	255.255.255.0	192.168.0.1	-
3	Laptop-Pt	192.168.0.102	Wireless0	255.255.255.0	192.168.0.1	-
4	Tablet-PC0	192.168.0.105	Wireless0	255.255.255.0	192.168.0.1	-
5	Printer-0		FastEthernet0	255.255.255.0	192.168.0.0	-

CONCLUSION:-

The experiment successfully demonstrates the configuration and deployment of a functional **Wireless Local Area Network using a WRT300N Wireless Router**. The experiment validates that a wireless infrastructure can replace traditional cabling while maintaining robust connectivity, security, and the ability to access external network resources through a centralized gateway.

EXPERIMENT NO: 7

Aim : Internetworking with routers: Design a three or four simple networks (with 3 to 4 hosts) and connect via Router. Perform simulation and trace how routing is done in packet transmission.

1: Experiment on same subnet

2: Perform Experiment across the subnet and observe functioning of Router via selecting suitable pair of Source and destination.

Purpose / Objective

- To understand the role of a **Router** in interconnecting multiple networks.
- To observe packet transmission when hosts are in the **same subnet**.
- To analyze how routing enables communication **across different subnets**.
- To trace the path of packets and verify routing table entries during simulation.

Logical Flow of Practical

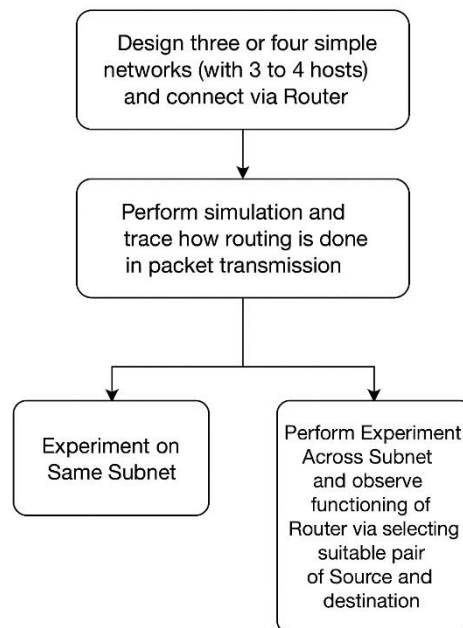
◆ Experiment 1: Communication in Same Subnet

1. Create **3–4 hosts** (PCs) in each of three small networks.
2. Connect each network to a **Router interface** using straight-through cables.
3. Assign IP addresses to all hosts **within the same subnet** (e.g., 192.168.1.x).
4. Configure router interfaces with IP addresses in the same subnet.
5. Run simulation: ping between hosts across different networks.
6. **Observation:** Since all networks are in the same subnet, routing is straightforward; packets are forwarded without subnet differentiation.

◆ Experiment 2: Communication Across Different Subnets

1. Create **3–4 hosts** in each of three small networks.
2. Connect each network to a **Router interface**.
3. Assign IP addresses to hosts in **different subnets** (e.g., 192.168.1.x, 192.168.2.x, 192.168.3.x).
4. Configure router interfaces with IP addresses corresponding to each subnet.

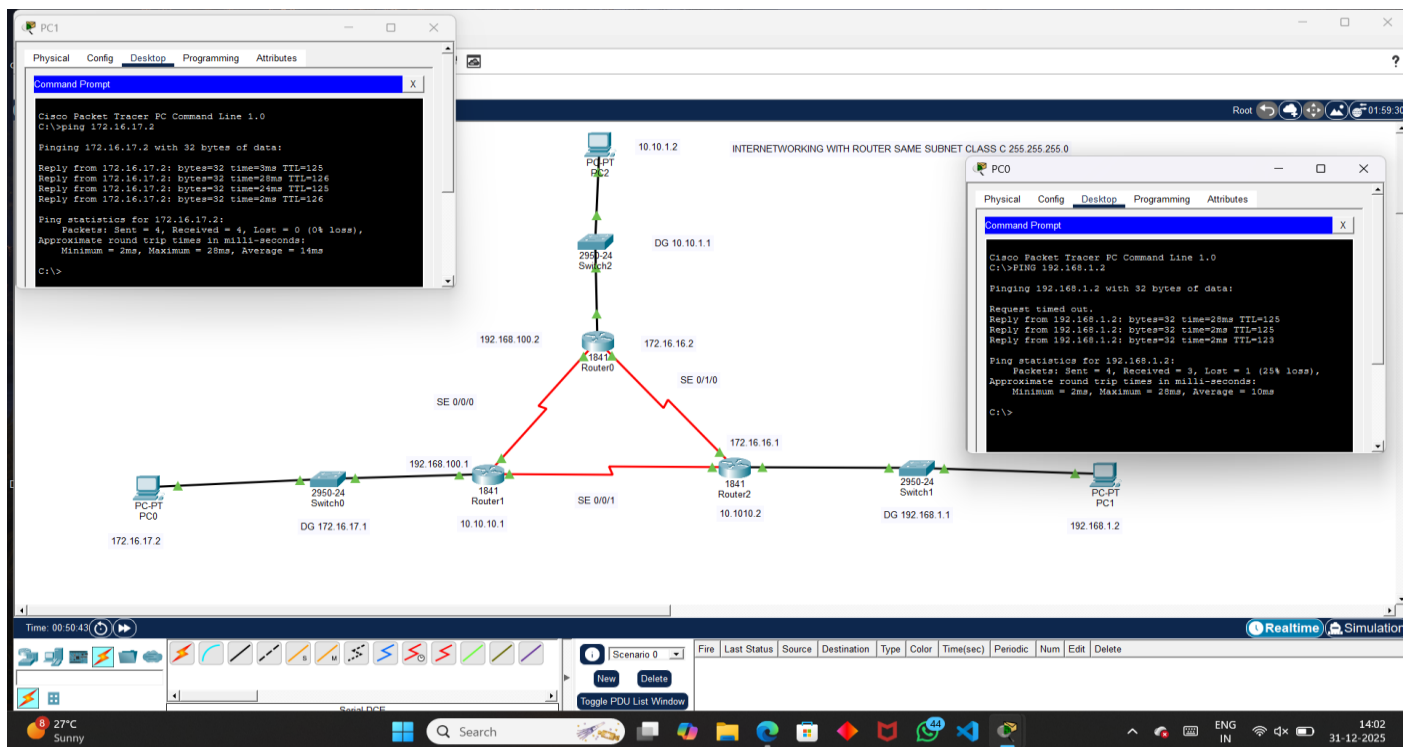
5. Run simulation: ping between hosts in different subnets (e.g., PC in 192.168.1.x → PC in 192.168.3.x).
6. **Observation:** Router checks its **routing table** and forwards packets to the correct subnet. Communication succeeds only because the router performs subnet-based routing.



Description

In this practical, multiple small networks are designed and interconnected using a **Router** in Cisco Packet Tracer.

- In the **same subnet experiment**, all hosts share a common subnet, and the router simply forwards packets between interfaces without complex routing decisions.
- In the **across subnet experiment**, hosts belong to different subnets, and the router uses its routing table to determine the correct path for packet delivery.
- By tracing packets in simulation mode, the difference in communication behavior is observed:
- **Same subnet:** Direct forwarding.
- **Different subnets:** Routing decisions based on subnet addresses.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	172.16.17.2	f0/1	255.255.255.0	192.168.1.1	0.0.0.0
2	PC1	192.168.1.2	f0/2	255.255.255.0	172.16.17.1	0.0.0.0
3	PC2	10.10.10.2	f0/3	255.255.255.0	10.10.10.1	0.0.0.0
4	2960-24TT Switch0	-	Router1	-	-	-
5	2960-24TT Switch1	-	Router2	-	-	-
6	2960-24TT Switch2	-	Router0	-	-	-
7	Router0(1841)	192.168.1.2	Se0/0/0	255.255.255.0	-	-
		172.16.16.2	Se0/1/0	255.255.255.0	-	-
8	Router1(1841)	192.168.1.1	Se0/0/0	255.255.255.0	-	-
		10.10.10.1	Se0/0/1	255.255.255.0	-	-
9	Router2(1841)	10.10.10.2	Se0/0/1	255.255.255.0	-	-
		172.16.16.1	Se0/1/0	255.255.255.0	-	-

CONCLUSION:

The experiment demonstrated that routers enable communication both within the same subnet and across different subnets. Routing tables ensure correct packet forwarding, proving the router's essential role in internetworking which was done successfully.

EXPERIMENT NO: 8

AIM- **Implementation of SUBNETTING:** Design multiple subnet with suitable number of hosts. Make a plan to assign static IP addressing across all subnet to explain implementation of SUBNETTING.

Purpose / Objective

- To understand how subnetting divides a large network into smaller, manageable sub-networks.
- To calculate subnet ranges and assign static IP addresses based on host requirements.
- To configure default gateways and verify inter-subnet communication.
- To simulate and observe packet transmission between devices in different subnets.

Logical Flow of Practical

1. Network Design

- Create two subnets: SUBNET#1 and SUBNET#2.
- Each subnet contains 3 PCs connected to a switch.
- Both switches connect to a central router (Router0).

2. IP Addressing Plan

- Use network 192.168.10.0/24 and divide it using subnet mask 255.255.255.128 (/25).
- SUBNET#1:
 - Network range: 192.168.10.0 – 192.168.10.127
 - PCs: 192.168.10.1, 192.168.10.2, 192.168.10.3
 - Default Gateway: 192.168.10.4
- SUBNET#2:
 - Network range: 192.168.10.128 – 192.168.10.255
 - PCs: 192.168.10.129, 192.168.10.130, 192.168.10.131
 - Default Gateway: 192.168.10.132

3. Static IP Configuration

- Assign IP addresses manually to each PC.
- Configure router interfaces with the respective default gateway IPs.
- Set subnet mask to 255.255.255.128 for all devices.

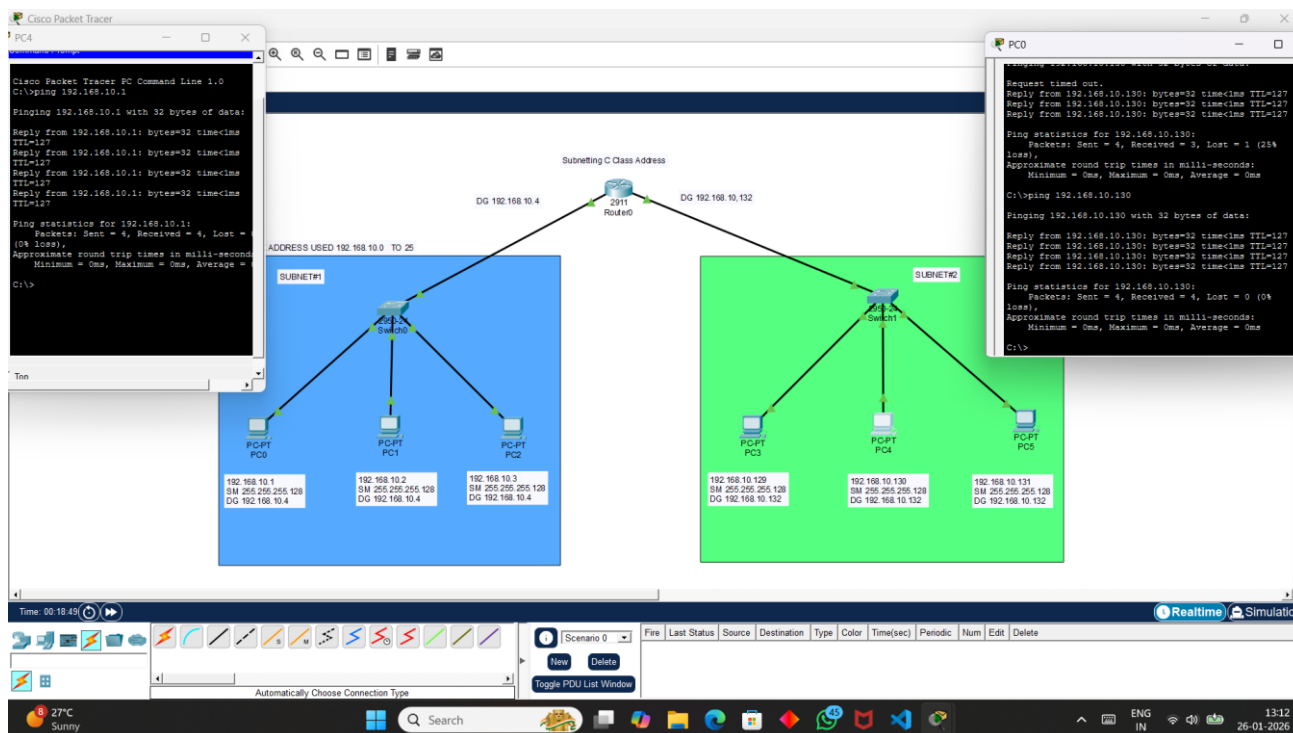
4. Simulation and Testing

- Use ping command to test connectivity between PCs in different subnets.
- Observe successful communication and TTL values.
- Verify that routing between subnets is handled correctly by the router.

Description of the Practical

This experiment demonstrates the implementation of **subnetting** by dividing a single network into two subnets using a /25 subnet mask. Each subnet is assigned a unique IP range and default gateway. Static IP addressing is used for all devices. The router acts as the intermediary for inter-subnet communication.

Ping tests confirm that devices in different subnets can communicate successfully, validating the correct setup of subnetting and routing.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	192.168.10.1	f0/1	255.255.255.128	192.168.10.4	-
2	PC1	192.168.10.2	f0/2	255.255.255.128	192.168.10.4	-
3	PC2	192.168.10.3	f0/3	255.255.255.128	192.168.10.4	-
4	PC3	192.168.10.129	f0/4	255.255.255.128	192.168.10.134	-
5	PC4	192.168.10.130	f0/5	255.255.255.128	192.168.10.134	-
6	PC5	192.168.10.131	f0/6	255.255.255.128	192.168.10.134	-
9	2960-24TT Switch0	-	Router0	-	-	-
10	2960-24TT Switch1	-	Router0	-	-	-
12	Router0	192.168.10.4	Gig0/0	255.255.255.128	-	-
		192.168.10.132	Gig0/1	255.255.255.128	-	-

CONCLUSION:

Subnetting was successfully implemented by dividing a single network into two subnets with static IP addressing. The experiment demonstrated efficient IP utilization and verified successful communication between subnets through a central router.

EXPERIMENT NO: 9

AIM - Routing at Network Layer:

Simulate Static and Dynamic Routing Protocol Configuration using CISCO Packet Tracer.

Purpose / Objective

- To understand the concept of **routing at the network layer**.
- To configure and test **Static Routing** between routers.
- To configure and test **Dynamic Routing Protocols** (e.g., RIP, OSPF, EIGRP) for automatic route learning.
- To compare the behavior of static and dynamic routing in packet transmission.

Logical Flow of Practical

1. Design a network with **3–4 routers**, each connected to a small LAN (with 2–3 PCs).
2. Assign IP addresses to all router interfaces and hosts.
3. Configure **static routes** manually on each router using the ip route command.
4. Run simulation: ping between hosts in different networks.
5. **Observation:** Communication succeeds only if correct static routes are configured. Any missing route causes failure.

Description

In this practical, **default routing** is configured in Cisco Packet Tracer to enable communication between multiple networks.

- A **default route** acts as a “catch-all” path for packets whose destination networks are not explicitly listed in the routing table.
- When a PC sends data to another subnet, the router forwards it to the next-hop router using the default route.
- This reduces the need for multiple static routes, especially in networks with a single exit point.

By simulating packet transmission, it is observed that communication across networks succeeds because routers rely on the default route to forward packets.

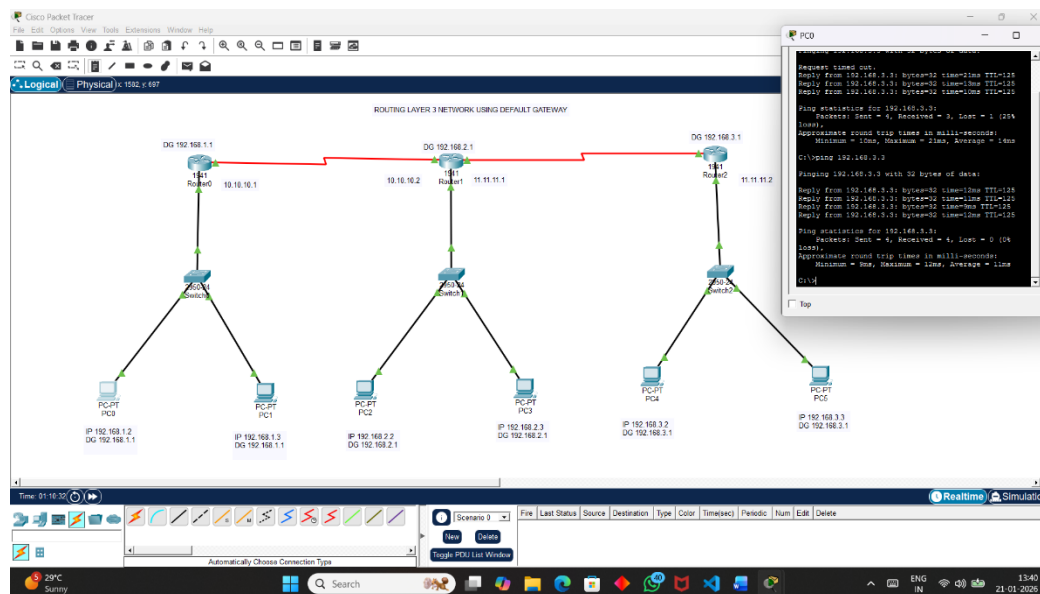
Default Route CLI Command

```
Router(config)# ip route 0.0.0.0 0.0.0.0 Serial0/0/0
```

```
To same as ) ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

Explanation:

- 0.0.0.0 0.0.0.0 → This defines the **default destination** (any unknown network).
- Serial0/1/0 → This is the **exit interface** through which packets will be forwarded.



S.No	Name of Components/Device	IP Address	Interface	Subnet Mask	Default Gateway IP	DNS Server IP
1	PC0	192.168.1.2	f0/1	255.255.255.0	192.168.1.1	-
2	PC1	192.168.1.3	f0/2	255.255.255.0	192.168.1.1	-
3	PC2	192.168.2.2	f0/3	255.255.255.0	192.168.2.1	-
4	PC3	192.168.2.3	f0/4	255.255.255.0	192.168.2.1	-
5	PC4	192.168.3.2	f0/5	255.255.255.0	192.168.3.1	-
6	PC5	192.168.3.3	f0/6	255.255.255.0	192.168.3.1	-
9	2960-24TT Switch0	-	Router0	-	-	-
10	2960-24TT Switch1	-	Router1	-	-	-
11	2960-24TT Switch2	-	Router2	-	-	-
12	Router0	192.168.1.1	Gig0/0	255.255.255.0	-	-
		10.10.10.1	Se0/0/0	255.0.0.0	-	-
13	Router1	192.168.2.1	Gig0/0	255.255.255.0	-	-
		10.10.10.2	Se0/0/0	255.0.0.0	-	-
		11.11.11.1	Se0/1/0	255.0.0.0	-	-
14	Router2	192.168.3.1	Gig0/0	255.255.255.0	-	-
		11.11.11.2	Se0/1/0	255.0.0.0	-	-

CONCLUSION:

In this experiment, we configured and simulated default routing at the network layer using Cisco Packet Tracer. The default route acted as a catch-all path, allowing routers to forward packets to unknown destinations. This demonstrated that default routing simplifies configuration and ensures successful communication across multiple networks.